

TECHNICAL REPORT

SMART ENVIRONMENTAL POLLUTION COMPLAINT AND MONITORING SYSTEM USING GEOTAGGING AND PRIORITY-BASED REPORTING



Namitha Thomas

Rollno:57

UID:U2503199

S3 CSE GAMMA

CONTENTS

SI NO	CONTENT	PAGE NO
1	Abstract	2
2	Introduction	3
3	Problem statement and Objectives	4
4	Sustainability analysis	5
5	Proposed Engineering Solution	9
6	Feasibility Analysis	16
7	Expected Environmental Benefits	18
8	Justification of the Proposed Solution	19
9	Future Scope	20
10	Conclusion	21
11	References	22

ABSTRACT

Environmental pollution is one of the major challenges affecting human health, ecosystems and the quality of life in communities. Pollution can occur in various forms, including air pollution, water pollution, land pollution, noise pollution and improper waste disposal. Although people frequently observe such problems in their surroundings, reporting them through conventional methods can be inconvenient, time-consuming and difficult to track.

This report proposes a Smart Environmental Pollution Complaint and Monitoring System, a responsive web-based platform that enables citizens to report environmental pollution easily. The proposed system allows users to select the type of pollution, provide a description, upload photographic evidence and submit the geographical location of the incident. Each complaint is assigned a unique complaint ID, allowing users to track its progress.

A key feature of the proposed solution is a priority-based complaint classification system. Complaints can be classified according to factors such as pollution severity, number of people affected, environmental sensitivity of the location and availability of supporting evidence. This allows authorities to identify serious complaints and prioritize them for further action.

The system supports sustainable environmental management by encouraging citizen participation, improving environmental data collection and providing a transparent mechanism for monitoring complaints. It also supports relevant Sustainable Development Goals, particularly SDG 15 – Life on Land, along with SDGs related to clean water, sustainable communities and responsible consumption.

INTRODUCTION

Environmental pollution refers to the introduction of harmful substances or forms of energy into the environment, causing adverse effects on living organisms, ecosystems and natural resources. Rapid urbanization, industrialization, population growth, improper waste disposal, excessive use of vehicles and unsustainable human activities have increased the occurrence of environmental pollution.

Pollution can affect air, water and land and can also create excessive noise in residential and public areas. Improper disposal of solid waste can contaminate soil and water bodies, while industrial and domestic wastewater can negatively affect aquatic ecosystems. Similarly, excessive air and noise pollution can affect the health and well-being of people living in urban areas.

An important step in controlling pollution is the early identification and reporting of environmental problems. Citizens are often the first people to observe issues such as illegal dumping, polluted water bodies, smoke emissions, overflowing waste, burning of waste and excessive noise. However, existing reporting methods may not always provide an easy way to submit evidence, identify the exact location and track the progress of a complaint.

Information Technology can provide an effective solution to this problem. A responsive web-based reporting platform can connect citizens and authorities through a centralized system. Users can submit complaints from smartphones or computers, while authorities can organize, prioritize and monitor complaints through an administrative dashboard.

The proposed Smart Environmental Pollution Complaint and Monitoring System focuses on making environmental reporting easier, more transparent and more organized. The system combines pollution reporting, geolocation, photographic evidence, complaint tracking and priority classification into one platform.

PROBLEM STATEMENT

Environmental pollution complaints are often difficult to report, verify and track efficiently. Citizens may not know where to report an issue, while authorities may receive many complaints without an effective way to prioritize serious cases.

Problem Statement:

There is a need for a centralized web-based system that allows citizens to report environmental pollution with location and photographic evidence while helping authorities prioritize, monitor and resolve complaints

OBJECTIVES

The main objective of the proposed system is to develop a responsive web-based platform for efficient environmental pollution reporting and monitoring.

The specific objectives are:

- To provide citizens with an easy platform for reporting pollution.
- To allow users to select different categories of pollution.
- To enable users to upload photographic evidence.
- To record the geographical location of pollution incidents.
- To generate a unique complaint ID for every report.
- To provide complaint status tracking.
- To classify complaints based on severity.
- To prioritize serious pollution incidents.
- To provide authorities with a centralized dashboard.
- To identify areas with repeated pollution complaints.
- To improve communication between citizens and authorities.
- To encourage public participation in environmental protection.

SUSTAINABILITY ANALYSIS

Relevant Sustainable Development Goals

The proposed system contributes to several Sustainable Development Goals.

SDG 15 – Life on Land

SDG 15: Life on Land focuses on protecting terrestrial ecosystems, forests, biodiversity and land resources.

The proposed system supports this goal by allowing citizens to report:

- Illegal dumping
- Land pollution
- Waste accumulation
- Damage to natural areas
- Pollution affecting terrestrial ecosystems
- Improper disposal of hazardous materials

Early reporting can help authorities identify environmental problems before they become more severe.

SDG 11 – Sustainable Cities and Communities

The system contributes to sustainable communities by allowing citizens to report pollution in residential and public areas.

It can support cleaner and safer communities by improving the identification and monitoring of environmental issues.

SDG 6 – Clean Water and Sanitation

Users can report:

- Water pollution
- Waste dumped into rivers
- Polluted ponds

- Sewage discharge
- Industrial waste entering water bodies

Such reports can help identify potential threats to water resources.

SDG 12 – Responsible Consumption and Production

Improper waste disposal is one of the environmental problems that can be reported through the system. The platform can encourage responsible waste management by making it easier to report illegal dumping and waste accumulation.

SDG 13 – Climate Action

Although the system does not directly reduce greenhouse gas emissions, it can improve environmental awareness and support monitoring of pollution-related problems that contribute to environmental degradation.

ISO 14001 ANALYSIS

ISO 14001 is an international standard that provides a framework for organizations to establish and improve an Environmental Management System (EMS).

The proposed website can support several principles associated with environmental management.

Environmental Problem Identification

The system provides a mechanism for identifying environmental incidents reported by citizens.

Documentation

Every complaint can be stored with information such as:

- Complaint ID
- Pollution type
- Description
- Date and time

- Location
- Photograph
- Severity
- Status

This creates an organized record of environmental incidents.

Monitoring

Authorities can monitor complaints through an administrative dashboard.

Corrective Action

Once a complaint is verified, authorities can assign it for corrective action and update its status.

Continual Improvement

Historical complaint data can help authorities identify frequently affected areas and improve environmental management strategies.

Therefore, the proposed system can act as a digital supporting tool for environmental management and monitoring.

LIFE CYCLE ANALYSIS

Life Cycle Analysis (LCA) evaluates the environmental impacts associated with a product or system throughout its life cycle.

For the proposed website, the major stages include:

System Development → Hosting → Data Storage → User Access → Maintenance → System Upgrading/Disposal

The system can have environmental benefits compared with paper-based reporting because information can be submitted and stored digitally.

However, digital systems also require electronic devices, servers, electricity and data storage. Therefore, environmentally responsible hosting and efficient data management

should be considered.

Possible measures include:

- Using energy-efficient servers
- Avoiding unnecessary data storage
- Compressing uploaded images
- Removing outdated or duplicate data
- Using efficient website design
- Encouraging digital rather than paper-based reporting

ENVIRONMENTAL IMPACT ASSESSMENT

A conventional Environmental Impact Assessment is generally associated with physical projects that may significantly affect the environment. The proposed website itself does not involve major physical construction or direct environmental disturbance.

However, the concept of environmental impact assessment can be considered in terms of the system's indirect benefits.

The collected complaint information can help identify:

- Pollution hotspots
- Frequently affected areas
- Waste accumulation zones
- Water pollution locations
- Areas requiring environmental intervention

Therefore, the website can provide useful information that may support environmental planning and decision-making.

PROPOSED ENGINEERING/IT SOLUTION

Overview

The proposed solution is a Smart Environmental Pollution Complaint and Monitoring System implemented as a responsive website.

The system consists of two primary users:

Citizen

Citizens can:

- Register/login
- Report pollution
- Upload photographs
- Provide location
- Track complaints
- View previous complaints

Authority/Administrator

Authorities can:

- View complaints
- Verify complaints
- Check evidence
- View locations
- Assign priority
- Update complaint status
- Monitor resolved and pending complaints
- Identify pollution hotspots

Working of the Proposed System

The basic working process is:

Citizen



Login/Register



Select Pollution Type



Enter Description



Upload Photograph



Provide/Detect Location



Submit Complaint



Generate Complaint ID



Priority Classification



Authority Dashboard



Verification and Action

↓

Update Complaint Status

↓

Citizen Tracks Complaint

This creates a complete reporting and monitoring cycle.

Pollution Categories

The system can provide different categories for users to select.

Air Pollution

Examples:

- Excessive smoke
- Industrial emissions
- Open burning
- Dust pollution

Water Pollution

Examples:

- Waste dumped into water bodies
- Sewage discharge
- Industrial discharge
- Oil or chemical contamination

Land Pollution

Examples:

- Illegal dumping
- Garbage accumulation
- Plastic waste

- Hazardous waste disposal

Noise Pollution

Examples:

- Excessive loudspeakers
- Industrial noise
- Construction noise
- Unusually high noise levels

Other Environmental Issues

Users can report environmental issues that do not fall under the main categories.

Geotagging

One of the important features of the proposed system is geotagging.

When submitting a complaint, the user can provide the location of the incident.

The location can be obtained through:

- Browser location services
- GPS-enabled mobile devices
- Manual location selection

Geotagging helps authorities determine exactly where the pollution incident has occurred.

It can also help identify areas where multiple complaints occur.

For example:

If several water pollution complaints are reported near the same river section, the authority can identify the area as a potential pollution hotspot.

Photographic Evidence

Users can upload photographs along with their complaints.

For example, a user reporting illegal dumping can upload a photograph showing the accumulated waste.

Photographic evidence can help authorities:

- Understand the problem
- Verify the complaint
- Estimate severity
- Identify the type of pollution
- Support further investigation

Priority-Based Complaint Classification

This is the unique feature of the proposed solution. Instead of treating every complaint equally, the system can classify complaints according to priority.

The priority can be determined using factors such as:

- Pollution severity
- Number of people affected
- Environmental sensitivity
- Location
- Supporting evidence

A simplified priority model can be represented as:

Priority Score = Severity + Population Impact + Environmental Sensitivity + Evidence

Based on the resulting score, complaints can be classified as:

Low Priority

Minor environmental issue with limited impact.

Medium Priority

Environmental issue affecting a local area.

High Priority

Serious pollution affecting people, water resources or ecosystems.

Critical Priority

Severe pollution requiring immediate attention.

This classification helps authorities focus their resources on the most serious cases.

Complaint Tracking

After submitting a complaint, the system generates a unique complaint ID.

Example:

Complaint ID: PCR-2026-00452

The user can use this ID to track the complaint.

Possible statuses include:

Submitted

↓

Under Review

↓

Assigned

↓

Action Taken

↓

Resolved

This improves transparency because citizens can know what stage their complaint has reached.

Authority Dashboard

The authority dashboard is designed to provide an overview of all complaints.

The dashboard can display:

- Total complaints
- Pending complaints
- Resolved complaints
- High-priority complaints
- Complaints by pollution type
- Complaints by location
- Recent complaints

Authorities can filter complaints according to:

- Pollution type
- Severity
- Date
- Location
- Status

This makes complaint management more organized.

Pollution Hotspot Identification

The geographical information collected from complaints can be used to identify areas where pollution occurs repeatedly.

For example:

If multiple complaints are received from the same location regarding waste dumping, the system can identify that area as a pollution hotspot.

This can help authorities move from simply responding to individual complaints toward identifying recurring environmental problems.

FEASIBILITY ANALYSIS

Technical Feasibility

The proposed system is technically feasible because it can be developed using commonly available web development technologies.

HTML, CSS, JavaScript, Bootstrap and database technologies can be used to create the system.

The website can also be designed to work on mobile devices, which makes it suitable for citizens reporting incidents from the field.

Economic Feasibility

The proposed system can be developed using open-source software and commonly available development tools.

A basic implementation does not require expensive hardware.

The main expenses could include:

- Web hosting
- Domain registration
- Database hosting
- Map services if paid APIs are used
- Maintenance

The system can therefore be developed at relatively low cost compared with larger physical environmental monitoring systems.

Operational Feasibility

The system is operationally feasible because it is designed to be simple for citizens.

A user only needs to:

1. Log in
2. Select the pollution type
3. Describe the problem

4. Upload evidence
5. Provide the location
6. Submit the complaint

Authorities can manage complaints through the dashboard.

Environmental Feasibility

The proposed system has the potential to reduce dependence on paper-based complaint forms.

It also supports early identification and monitoring of environmental problems.

However, the website itself consumes electricity through user devices, servers and data storage. Therefore, efficient hosting and responsible data management should be considered.

EXPECTED ENVIRONMENTAL BENEFITS

The proposed system can provide several environmental and social benefits.

Faster Pollution Reporting

Citizens can report environmental issues immediately instead of relying only on conventional reporting methods.

Improved Documentation

Photographs, descriptions, locations and timestamps create better documentation of pollution incidents.

Better Environmental Monitoring

Authorities can view complaints in a centralized system.

Identification of Pollution Hotspots

Repeated complaints from the same area can help identify locations requiring additional attention.

Increased Public Participation

The platform encourages citizens to actively participate in environmental protection.

Better Prioritization

Serious pollution complaints can receive higher priority.

Improved Transparency

Complaint tracking allows users to monitor the progress of their reports.

Reduced Paper Usage

Digital reporting can reduce the need for physical complaint forms.

JUSTIFICATION OF THE PROPOSED SOLUTION

The proposed system is justified because environmental pollution cannot be addressed only through government action. Citizens who live and work in affected areas can provide valuable information about pollution incidents.

A centralized digital platform creates a connection between citizens and authorities.

The proposed solution is particularly useful because it combines several features in a single platform:

- Pollution reporting
- Photographic evidence
- Geotagging
- Priority classification
- Complaint tracking
- Authority dashboard
- Pollution hotspot identification

The priority-based classification system makes the solution different from a basic complaint form. Instead of simply collecting complaints, the system attempts to organize them according to their potential environmental and social impact.

For example, a complaint about minor waste accumulation and a complaint about industrial waste entering a drinking-water source should not necessarily receive the same priority. The proposed system provides a mechanism for distinguishing such cases.

The solution is also scalable. Future versions could include additional features such as artificial intelligence for image-based pollution classification, real-time environmental sensor data and mobile applications.

Therefore, the proposed system provides a practical and scalable IT-based approach to environmental pollution reporting and monitoring.

FUTURE SCOPE

The proposed system can be improved in several ways in the future.

Artificial Intelligence

AI can be used to analyze uploaded photographs and identify possible types of pollution.

For example, an image could potentially be analyzed to identify:

- Plastic waste
- Garbage accumulation
- Smoke
- Water contamination indicators

Real-Time Sensors

IoT sensors could be integrated to collect:

- Air quality
- Noise levels
- Water quality
- Temperature
- Other environmental parameters

Mobile Application

A dedicated Android/iOS application could provide faster reporting.

Automatic Pollution Hotspot Detection

Machine learning could be used to analyze historical complaint data and identify areas with increasing pollution incidents.

Multilingual Support

The website could support multiple regional languages to make it accessible to more citizens.

CONCLUSION

Environmental pollution is a serious problem that requires effective monitoring, public participation and timely corrective action. Citizens often observe pollution incidents in their surroundings, but conventional reporting methods may not always provide an efficient mechanism for submitting evidence, identifying locations and tracking complaints.

The proposed Smart Environmental Pollution Complaint and Monitoring System provides an IT-based solution to these challenges. The responsive website allows users to report different types of pollution, upload photographic evidence, provide geographical information and track the status of their complaints.

The most important feature of the proposed solution is its priority-based complaint classification system, which can help authorities identify serious pollution incidents and allocate attention accordingly. Geotagging and historical complaint information can also support the identification of pollution hotspots.

The proposed system contributes to sustainable environmental management and supports SDG 15 – Life on Land, along with SDGs related to sustainable communities, clean water and responsible consumption. It can also support environmental management practices associated with ISO 14001.

Although the website cannot directly eliminate pollution, it can serve as an effective tool for reporting, monitoring, prioritizing and managing environmental problems. With future integration of artificial intelligence, IoT sensors, data analytics and mobile applications, the system can become a more comprehensive environmental monitoring platform.

Thus, the proposed solution demonstrates how Engineering and Information Technology can be applied to encourage citizen participation and support more efficient environmental protection.

REFERENCES

1. United Nations. Sustainable Development Goals. United Nations.
2. United Nations. Sustainable Development Goal 15: Life on Land. United Nations.
3. International Organization for Standardization. ISO 14001 – Environmental Management Systems.
4. United Nations Environment Programme (UNEP). Environmental Pollution and Environmental Protection Resources.
5. World Health Organization (WHO). Air Pollution and Environmental Health Resources.
6. International Organization for Standardization. ISO 14040 – Environmental Management – Life Cycle Assessment – Principles and Framework.
7. International Organization for Standardization. ISO 14044 – Environmental Management – Life Cycle Assessment – Requirements and Guidelines.
8. World Bank. Environmental Management and Pollution-related Resources.